Andy's Package

Command cheat sheet

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The printed column shows the symbol. The two source columns show how to type it with and without the package. Commands marked as optional need their named package option. For an LLM transcription prompt, copy `docs/llm-prompt.txt`; it is printed in the quick-start guide.

Delimiters

Printed	With package	Without package	Use
(x)	`\parens {x}`	`\left (x\right )`	parentheses
$[x]$	`\bracks {x}`	`\left [x\right ]`	square brackets
$\{x\}$	`\braces {x}`	`\left \{x\right \}`	braces
$\langle x\rangle$	`\angles {x}`	`\left \langle x\right \rangle`	angle brackets
$\lvert x\rvert$	`\abs {x}`	`\lvert x\rvert`	absolute value
$\lVert x\rVert$	`\norm {x}`	`\lVert x\rVert`	norm
$\lceil x\rceil$	`\ceil {x}`	`\left \lceil x\right \rceil`	ceiling
$\lfloor x\rfloor$	`\floor {x}`	`\left \lfloor x\right \rfloor`	floor
$\{x,y\}$	`\set {x,y}`	`\left \{x,y\right \}`	set

Logic

Printed	With package	Without package	Use
$\vdash$	`\proves`	`\vdash`	syntactic derivability
$\nvdash$	`\notproves`	`\nvdash`	non-derivability
$\vDash$	`\entails`	`\vDash`	semantic consequence
$\nvDash$	`\notentails`	`\nvDash`	non-entailment
$\models$	`\satisfies`	`\models`	truth in one structure
$\not\models$	`\notsatisfies`	`\not \models`	non-satisfaction
$\Vdash$	`\forces`	`\Vdash`	forcing
$\nVdash$	`\notforces`	`\nVdash`	non-forcing
$\bot$	`\contradiction`	`\bot`	contradiction
$\Longrightarrow\bot$	`\impliesbottom`	`\Longrightarrow \bot`	implication to bottom
$\Longrightarrow\bot$	`\impliesbot`	`\Longrightarrow \bot`	alias
$\bot$	`\bottom`	`\bot`	bottom
$\top$	`\yippee`	`\top`	top
$p \Vdash_P \varphi$	`\forcesP {P}{p}{\varphi }`	`p\Vdash _P\varphi`	forcing indexed by poset
$\Gamma \vdash_S \Delta$	`\turnstile [S]{\Gamma }{\Delta }`	`\Gamma \vdash _S\Delta`	labeled sequent
$\vdash$	`\fCenter`	`\vdash`	bussproofs center symbol

Metavariables

Printed	With package	Without package	Use
$\mathscr{V}$	`\Var`	`\mathcal {V}`	variables
$\mathscr{F}$	`\Frm`	`\mathcal {F}`	formulas
$\mathscr{M}$	`\Str`	`\mathcal {M}`	structure
$\mathscr{L}$	`\Lang`	`\mathcal {L}`	language
Th	`\Th`	`\mathrm {Th}`	theory

Maps and arrows

Printed	With package	Without package	Use
$\rightarrowtail$	`\inj`	`\rightarrowtail`	injection
$\twoheadrightarrow$	`\surj`	`\twoheadrightarrow`	surjection
$\hookrightarrow$	`\incl`	`\hookrightarrow`	inclusion map
$\not\hookrightarrow$	`\notincl`	`\not \hookrightarrow`	not inclusion map
$\xrightarrow{\sim}$	`\isoarrow`	`\xrightarrow {\sim }`	isomorphism arrow
$\leftrightarrow$	`\bij`	`\leftrightarrow`	bijection-exists convention
$\rightarrowtail\!\!\!\!\twoheadrightarrow$	`\isbij`	`\rightarrowtail \mkern -14.75mu\twoheadrightarrow`	composite bijection arrow
$\Longleftrightarrow$	`\iffarrow`	`\Longleftrightarrow`	logical iff
$\hookrightarrow\!\!\!\!\twoheadrightarrow$	`\barbbij`	`\mathrlap {\hookrightarrow }\mkern 4.5mu\twoheadrightarrow`	barbed bijection
$\not\hookrightarrow\!\!\!\!\twoheadrightarrow$	`\notbarbbij`	`\not \!(\mathrlap {\hookrightarrow }\mkern 4.5mu\twoheadrightarrow )`	negated barbed bijection
$\leftrightarrow$	`\hbij`	`\rotatebox {180}{\hookrightarrow }\ \hookrightarrow`	hooked two-way arrow

Coding and vectors

Printed	With package	Without package	Use
$\ulcorner\varphi\urcorner$	`\gcode {\varphi }`	`\ulcorner \varphi \urcorner`	Gödel corners
$\ulcorner$	`\gcodel`	`\ulcorner`	left corner
$\urcorner$	`\gcoder`	`\urcorner`	right corner
$\llcorner\varphi\lrcorner$	`\gfloor {\varphi }`	`\llcorner \varphi \lrcorner`	lower corners
$\ulcorner\varphi\urcorner$	`\tarski {\varphi }`	`\ulcorner \varphi \urcorner`	coding alias
$\overrightarrow{v}$	`\harpvec {v}`	`\accentset {\rightharpoonup }{v}`	harpoon vector
$\overrightarrow{\mathbf{0}}$	`\zerovec`	`\accentset {\rightharpoonup }{\mathbf 0}`	zero vector

Kripke and structures

Printed	With package	Without package	Use
$\langle a,b\rangle$	`\tuple {a,b}`	`\langle a,b\rangle`	tuple
$\langle A,R\rangle$	`\struc {A,R}`	`\langle A,R\rangle`	structure
$\succeq$	`\kripkesucceq`	`\succeq`	accessibility orientation
$\succ$	`\kripkesucc`	`\succ`	strict accessibility
$\preceq$	`\kripkepreceq`	`\preceq`	converse
$\prec$	`\kripkeprec`	`\prec`	strict converse
$\mathscr{W}$	`\Worlds`	`\mathcal W`	worlds; kripke option
$\succeq$	`\Acc`	`\succeq`	legacy alias; kripke option
$\mathscr{F}$	`\Frame`	`\mathcal F`	frame; kripke option
$\mathscr{M}$	`\Model`	`\mathcal M`	model; kripke option
$\langle W,R\rangle$	`\kripkeframe {W}{R}`	`\langle W,R\rangle`	frame; kripke option
$\langle W,R,V\rangle$	`\kripkemodel {W}{R}{V}`	`\langle W,R,V\rangle`	model; kripke option
$\succ$	`\AccS`	`\succ`	strict legacy alias
$\preceq$	`\AccInv`	`\preceq`	converse legacy alias
$\prec$	`\AccInvS`	`\prec`	strict converse legacy alias
$\geqslant$	`\gecurly`	`\geqslant`	slanted greater-or-equal

Blackboard bold

Printed	With package	Without package	Use
$\mathbb{N}$	`\NN`	`\mathbb N`	naturals
$\mathbb{Z}$	`\ZZ`	`\mathbb Z`	integers
$\mathbb{Q}$	`\QQ`	`\mathbb Q`	rationals

Printed	With package	Without package	Use
$\mathbb{R}$	`\RR`	`\mathbb R`	reals
$\mathbb{C}$	`\CC`	`\mathbb C`	complex numbers
$\mathbb{F}$	`\FF`	`\mathbb F`	field
$\mathbb{P}$	`\PP`	`\mathbb P`	context-dependent P
$\mathbb{A}$	`\AAA`	`\mathbb A`	blackboard A
$\mathbb{B}$	`\BBB`	`\mathbb B`	blackboard B
$\mathbb{S}$	`\SSS`	`\mathbb S`	blackboard S
$\mathbb{X}$	`\XXX`	`\mathbb X`	blackboard X
$\mathbb{R}$	`\Reals`	`\mathbb R`	verbose alias
$\mathbb{N}$	`\Naturals`	`\mathbb N`	verbose alias
$\mathbb{Z}$	`\Integers`	`\mathbb Z`	verbose alias
$\mathbb{Q}$	`\Rationals`	`\mathbb Q`	verbose alias
$\mathbb{C}$	`\Complexes`	`\mathbb C`	verbose alias

Alphabets

Printed	With package	Without package	Use
$\mathscr{L}$	`\MathCal {L}`	`\mathcal L`	calligraphic
$\mathbb{R}$	`\MathBB {R}`	`\mathbb R`	blackboard bold
$\mathbf{v}$	`\MathBold {v}`	`\mathbf v`	bold
Th	`\MathRoman {Th}`	`\mathrm {Th}`	upright
$\mathscr{L}$	`\MathScript {L}`	`\mathscr L`	script; fallback calligraphic

Model and set theory

Printed	With package	Without package	Use
FOL	`\FOL`	`\mathrm {FOL}`	first-order logic
$f \upharpoonright_A$	`\restrict {f}{A}`	`f\!\upharpoonright _A`	restriction
dom	`\dom`	`\operatorname {dom}`	domain
ran	`\ran`	`\operatorname {ran}`	range
field	`\field`	`\operatorname {field}`	field
rank	`\rank`	`\operatorname {rank}`	rank
TC	`\TC`	`\operatorname {TC}`	transitive closure
cf	`\cf`	`\operatorname {cf}`	cofinality
otp	`\otp`	`\operatorname {otp}`	order type
Ord	`\Ord`	`\mathrm {Ord}`	ordinals
$\mathscr{P}(X)$	`\Pow {X}`	`\mathcal P\!\left (X\right )`	power set
Mod	`\Mod`	`\operatorname {Mod}`	models operator

Relations

Printed	With package	Without package	Use
$\subseteq$	`\containedin`	`\subseteq`	subset or equal
$\subset$	`\propcontainedin`	`\subset`	author-dependent proper subset
$\subsetneq$	`\emphpropcontainedin`	`\subsetneq`	unambiguous proper subset
$\nsubseteq$	`\notcontainedin`	`\nsubseteq`	not subset or equal
$\not\subset$	`\notpropcontainedin`	`\not \subset`	not proper subset
$\supseteq$	`\containedby`	`\supseteq`	superset or equal
$\supset$	`\propcontainedby`	`\supset`	author-dependent proper superset
$\supsetneq$	`\emphpropcontainedby`	`\supsetneq`	unambiguous proper superset
$\nsupseteq$	`\notcontainedby`	`\nsupseteq`	not superset or equal
$\not\supset$	`\notpropcontainedby`	`\not \supset`	not proper superset
$\cong$	`\isoto`	`\cong`	isomorphic
$\in$	`\belongsto`	`\in`	membership
$\wedge$	`\aand`	`\land`	conjunction
$\wedge$	`\conjunction`	`\land`	conjunction alias

Printed	With package	Without package	Use
$\vee$	`\oor`	`\lor`	disjunction
$\vee$	`\disjunction`	`\lor`	disjunction alias
$\nexists$	`\notexists`	`\nexists`	nonexistence

Proof trees

Printed	With package	Without package	Use
rulelabel	`\rlabel {Cut}`	`\RightLabel {\scriptsize Cut}`	rule label helper; prooftrees option
—	`\WeakL {A}`	`\RightLabel {\scriptsize (Weak L)}\UnaryInfC {A}`	weakening left
—	`\WeakR {A}`	`\RightLabel {\scriptsize (Weak R)}\UnaryInfC {A}`	weakening right
—	`\ContrL {A}`	`\RightLabel {\scriptsize (Contr L)}\UnaryInfC {A}`	contraction left
—	`\ContrR {A}`	`\RightLabel {\scriptsize (Contr R)}\UnaryInfC {A}`	contraction right
—	`\ExchL {A}`	`\RightLabel {\scriptsize (Exch L)}\UnaryInfC {A}`	exchange left
—	`\ExchR {A}`	`\RightLabel {\scriptsize (Exch R)}\UnaryInfC {A}`	exchange right
—	`\Cut {A}`	`\RightLabel {\scriptsize (Cut)}\BinaryInfC {A}`	cut
—	`\ImpR {A}`	`\RightLabel {\scriptsize ($\to $R)}\UnaryInfC {A}`	implication right
—	`\ImpL {A}`	`\RightLabel {\scriptsize ($\to $L)}\BinaryInfC {A}`	implication left
—	`\AndR {A}`	`\RightLabel {\scriptsize ($\wedge $R)}\BinaryInfC {A}`	and right
—	`\AndL {A}`	`\RightLabel {\scriptsize ($\wedge $L)}\UnaryInfC {A}`	and left
—	`\OrRone {A}`	`\RightLabel {\scriptsize ($\vee $R1)}\UnaryInfC {A}`	or right 1
—	`\OrRtwo {A}`	`\RightLabel {\scriptsize ($\vee $R2)}\UnaryInfC {A}`	or right 2
—	`\OrL {A}`	`\RightLabel {\scriptsize ($\vee $L)}\BinaryInfC {A}`	or left
—	`\NegR {A}`	`\RightLabel {\scriptsize ($\neg $R)}\UnaryInfC {A}`	negation right
—	`\NegL {A}`	`\RightLabel {\scriptsize ($\neg $L)}\UnaryInfC {A}`	negation left
—	`\TopR {A}`	`\RightLabel {\scriptsize ($\top $R)}\UnaryInfC {A}`	top right
—	`\BotL {A}`	`\RightLabel {\scriptsize ($\bot $L)}\UnaryInfC {A}`	bottom left
—	`\AllR {A}`	`\RightLabel {\scriptsize ($\forall $R)}\UnaryInfC {A}`	forall right
—	`\AllL {A}`	`\RightLabel {\scriptsize ($\forall $L)}\UnaryInfC {A}`	forall left
—	`\ExR {A}`	`\RightLabel {\scriptsize ($\exists $R)}\UnaryInfC {A}`	exists right
—	`\ExL {A}`	`\RightLabel {\scriptsize ($\exists $L)}\UnaryInfC {A}`	exists left

Document helpers

Printed	With package	Without package	Use
—	`\LogicLegend`	`\Gamma \vdash \varphi ; \Gamma \vDash \varphi`	insert relation legend
—	`\KripkeLegend`	`manual explanatory paragraph`	insert Kripke legend; kripke option
—	`\IJSection {Heading}`	`\section {Heading} plus legends`	section with optional legends

Formatting

Printed	With package	Without package	Use
—	`\AndyMakeTitle`	`manual center environment`	flexible inline title
—	`\AndyMakeTitlePage`	`titlepage environment`	separate title page
—	`\AndyStartBodyPages`	`\clearpage \pagenumbering {arabic}\setcounter {page}{0}`	body pagination
—	`\AndySubtitle {Text}`	`manual title text`	optional subtitle
—	`\AndyCourse {Text}`	`manual title text`	optional course
—	`\AndyAssignment {Text}`	`manual title text`	optional assignment
—	`\AndyDue {Text}`	`manual title text`	optional due date
—	`\AndyAffiliation {Text}`	`manual title text`	optional affiliation
—	`\AndyTitle []{A}{1}{Title}{Due}{Name}`	`manual center environment`	legacy six-argument title
WLOG	`\WLoG`	`\textsc {wlog}`	without loss of generality
WTS	`\WTS`	`\textsc {wts}`	want to show
WRT	`\WRT`	`\textsc {wrt}`	with respect to
SFSoC	`\SFSoC`	`\textsc {sfsoc}`	suppose for sake of contradiction
—	`\DearFriends {Name}{Text}`	`lettrine and calligra`	letter opening
—	`\AndyRunningTitle {Short title}`	`fancyhdr running head`	optional even-page short title
—	`\AndyStudentID {Text}`	`manual header text`	optional student ID; display with studentid=true
—	`\AndyBookMark [QED]`	`manual TikZ tombstone`	hollow: quoted from or closely follows a source
—	`\AndyGeneralMark [AXM]`	`manual TikZ tombstone`	striped: general proof; optional provenance tag
—	`\AndyOriginalMark [QED]`	`manual TikZ tombstone`	solid purple: highly original contribution
—	`\AndyProofMark [Tag]{style}`	`manual TikZ tombstone`	low-level hollow/general/solid proof mark
—	`\AndyCaseProgressMark {1}{3}`	`manual TikZ tombstone`	low-level completed/total case marker
—	`\AndyTombstoneFootnote`	`manual footnote`	short proof-mark and provenance explanation
—	`\AndyProofNotationAppendix`	`manual section text`	long proof-mark and provenance explanation
—	`\AndyOrnamentBreak`	`manual ornament and rules`	Angus-inspired decorative divider

Environments

Printed	With package	Without package	Use
—	`definition / DefinitionBlock`	`amsthm newtheorem / breakable tcolorbox`	shared subsection counter
—	`lemma / LemmaBlock`	`amsthm newtheorem / breakable tcolorbox`	shared subsection counter
—	`proposition / PropositionBlock`	`amsthm newtheorem / breakable tcolorbox`	shared subsection counter
—	`claim / ClaimBlock`	`amsthm newtheorem / breakable tcolorbox`	shared subsection counter
—	`example / ExampleBlock`	`amsthm newtheorem / breakable tcolorbox`	shared subsection counter
—	`chaff / ChaffBlock`	`amsthm newtheorem / breakable tcolorbox`	shared subsection counter
—	`aside / AsideBlock`	`amsthm newtheorem / breakable tcolorbox`	shared subsection counter
—	`theorem / TheoremBlock`	`amsthm newtheorem / breakable tcolorbox`	separate theorem counter
—	`conjecture / ConjectureBlock`	`amsthm newtheorem / breakable tcolorbox`	separate conjecture counter
—	`corollary / CorollaryBlock`	`amsthm newtheorem / breakable tcolorbox`	roman under theorem
—	`proof / ProofBlock`	`amsthm proof / thin separators`	page-break-friendly
—	`remark / RemarkBlock`	`amsthm newtheorem / breakable tcolorbox`	shared subsection counter
—	`note / NoteBlock`	`amsthm newtheorem / breakable tcolorbox`	shared subsection counter
—	`exercise / ExerciseBlock`	`amsthm newtheorem / breakable tcolorbox`	shared subsection counter
—	`ntheorem / nlemma / ndefinition`	`theorem / lemma / definition`	legacy aliases
—	`nproposition / ncorollary / nremark`	`proposition / corollary / remark`	legacy aliases
—	`nchaff / naside / nclaim`	`chaff / aside / claim`	legacy aliases
—	`IffForwardProof / IffBackwardProof`	`amsthm proof with custom qedsymbol`	biconditional directions; optional provenance tag

Printed	With package	Without package	Use
—	`CaseProof / AndyCase`	`manual case list with progress marks`	automatic “Proceed by cases” and progressive tombstones